

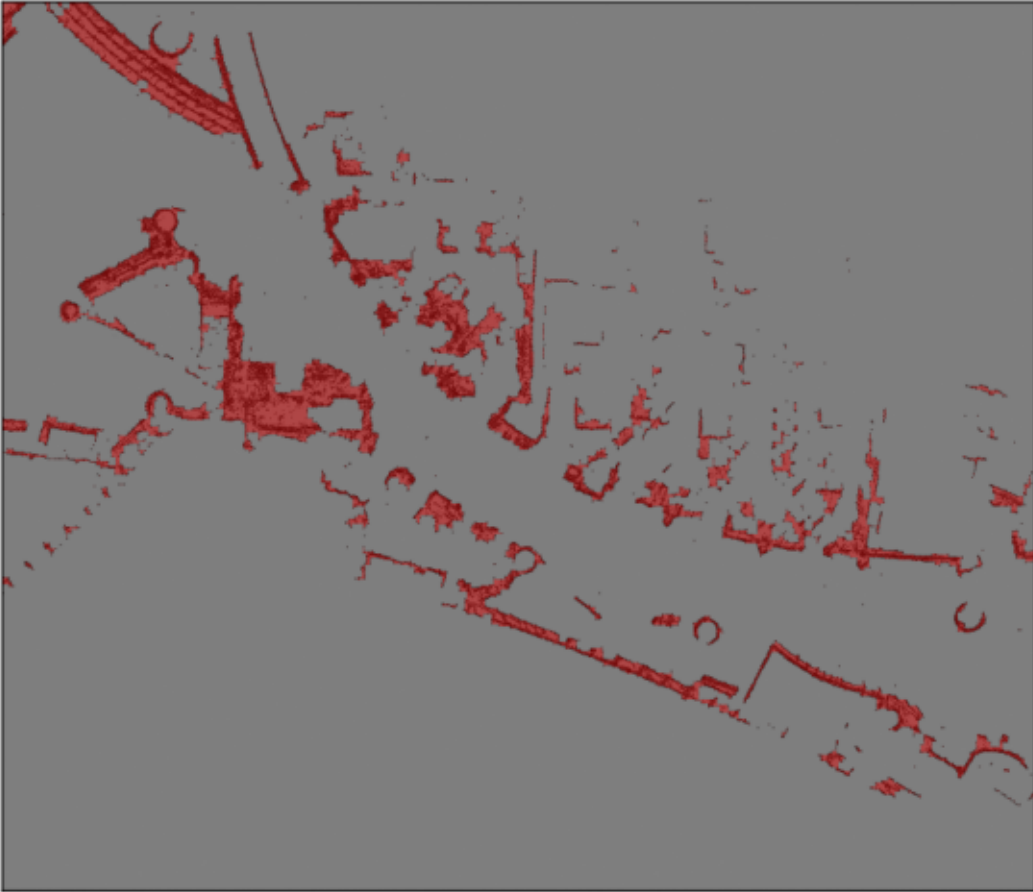
Progress: 50% map · full-data training · fair EnKF comparison

Reproducing 4DVarNet (Fablet 2020) on ATC crowd data — results for this meeting

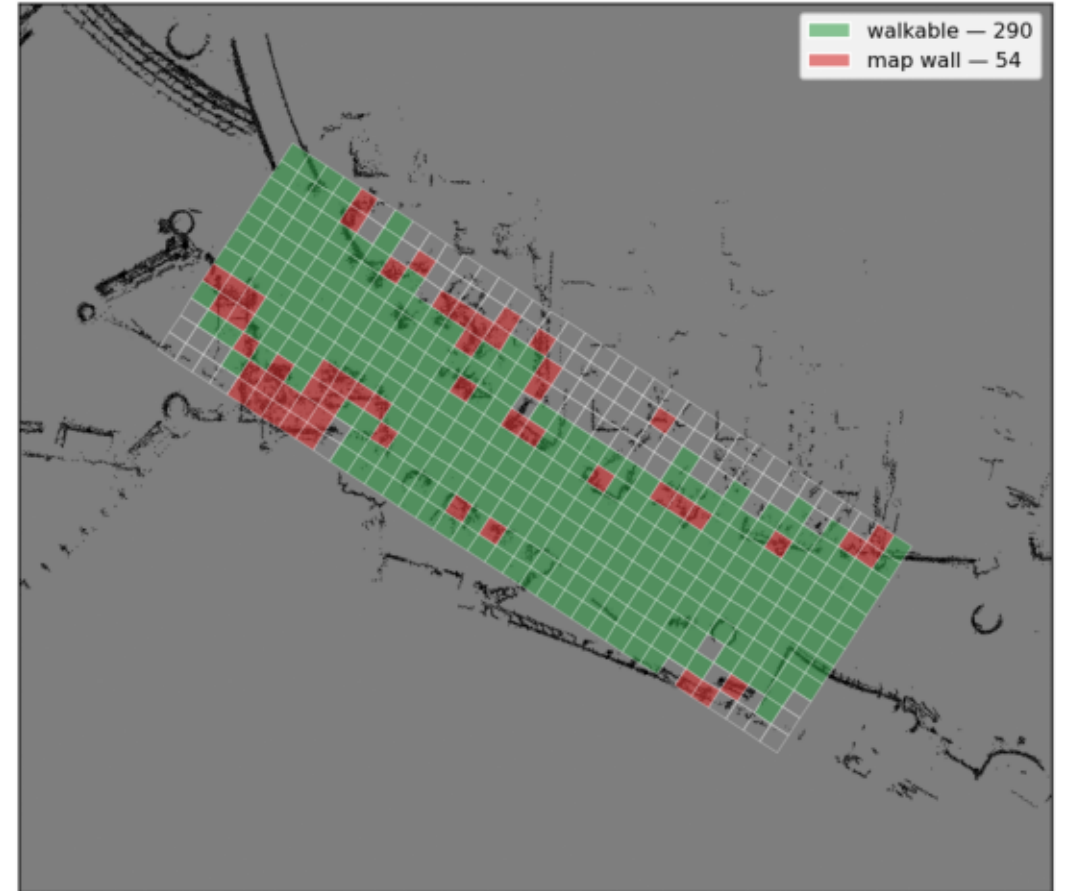
Xinle Zhang · July 2026

1 — Walkable map: supervisor's 50% occupancy rule

solid obstacle occupancy (ROS: OCCUPIED u enclosed-UNKNOWN)
the 50% rule is measured per 1 m cell against THIS mask (7% of the crop is obstacle)

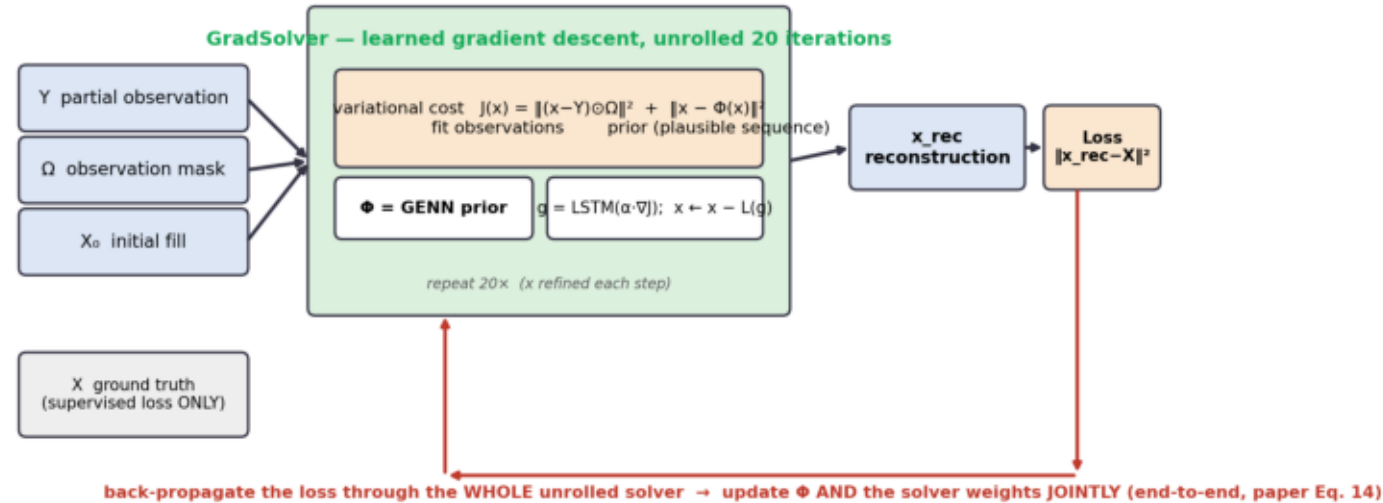


walkable mask (filled 1 m cells) — HYBRID: pedestrian ever present AND $\geq 50\%$ obstacle rule
290/432 walkable (1 m grid = state resolution, largest component)



- left: solid obstacle occupancy (ROS OCCUPIED u enclosed-UNKNOWN — pillar/stall interiors filled) = what the rule measures. right: resulting walkable (green) vs wall cells
- supervisor's 50% rule: a 1 m cell is WALL iff $\geq 50\%$ obstacle (config obstacle_rule='per_cell', occupancy_thresh=0.5) · + 'walked-in-training' hybrid · largest connected component

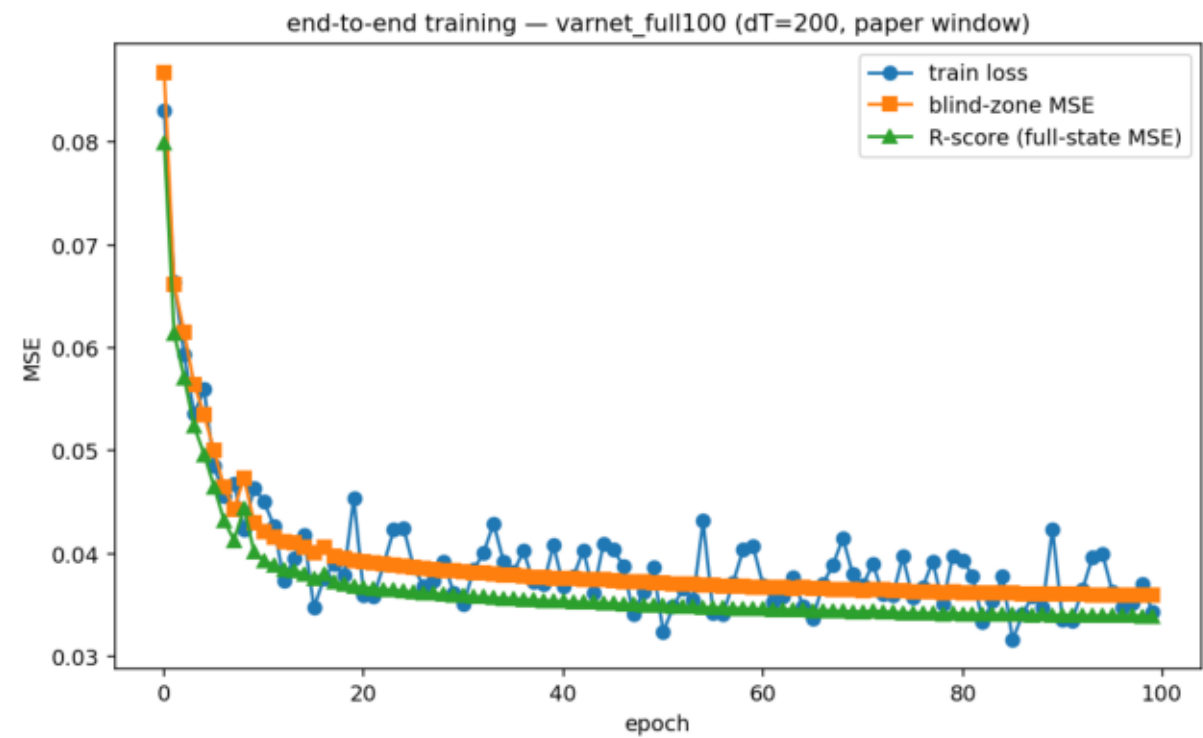
2 — The framework (recap): learned variational reconstruction



- reconstruct the FULL crowd field over a 200-frame window from robots' partial, noisy observations
- cost $J(x) = \|(x - y) \odot \Omega\|^2$ (fit what the robots saw) + $\|x - \Phi(x)\|^2$ (Φ scores dynamical plausibility)
- a learned gradient descent unrolls 20 steps from a rough fill X_0 ; step 20 = the reconstruction — uses ONLY observations + Φ , never the truth
- trained END-TO-END: the loss $\|x_{\text{rec}} - X\|^2$ back-props through the whole unrolled solver → Φ and the solver optimise jointly (paper Eq.14)

3 — Trained to convergence on the full 32-day dataset (dT=200)

setting	value
framework / loss	4DVarNet (Fablet 2020); supervised $\ x_{\text{rec}} - X\ ^2$ (Eq.14)
optimizer / lr	Adam / 0.001
epochs / batch / precision	100 / 32 / bf16 (amp)
window dT / solver iters	200 (paper window) / 20
total trainable params	2.05 M
data	32 days; windows of (4, 200, 36, 12)
observation	3 robots; range 7; line-of-sight; every 4th frame; +noise
hardware	1 × NVIDIA H200

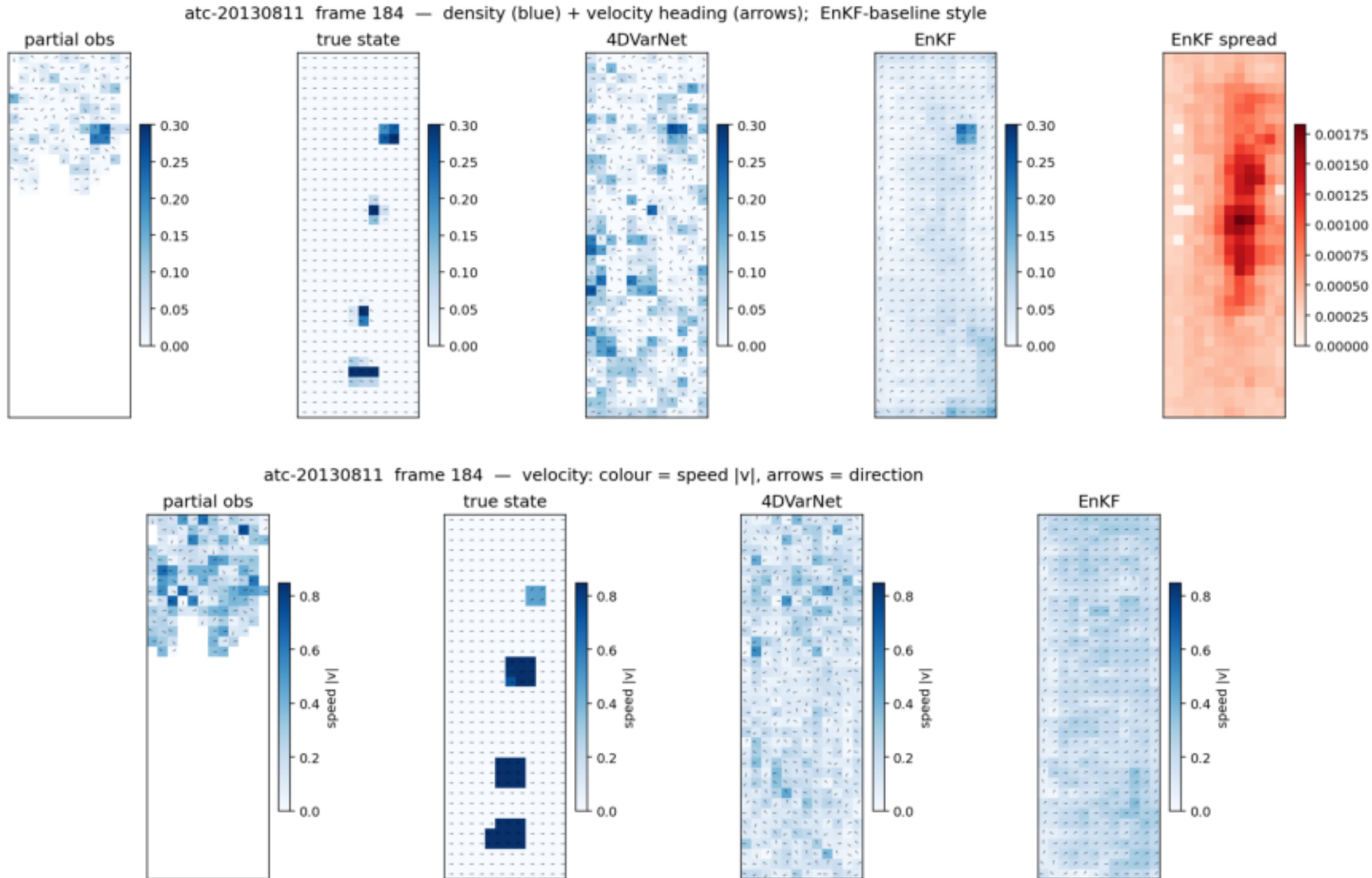


train loss · blind-zone MSE · R-score vs epoch (converges by ~epoch 20)

4 — Fair comparison: identical everything, no truth at init

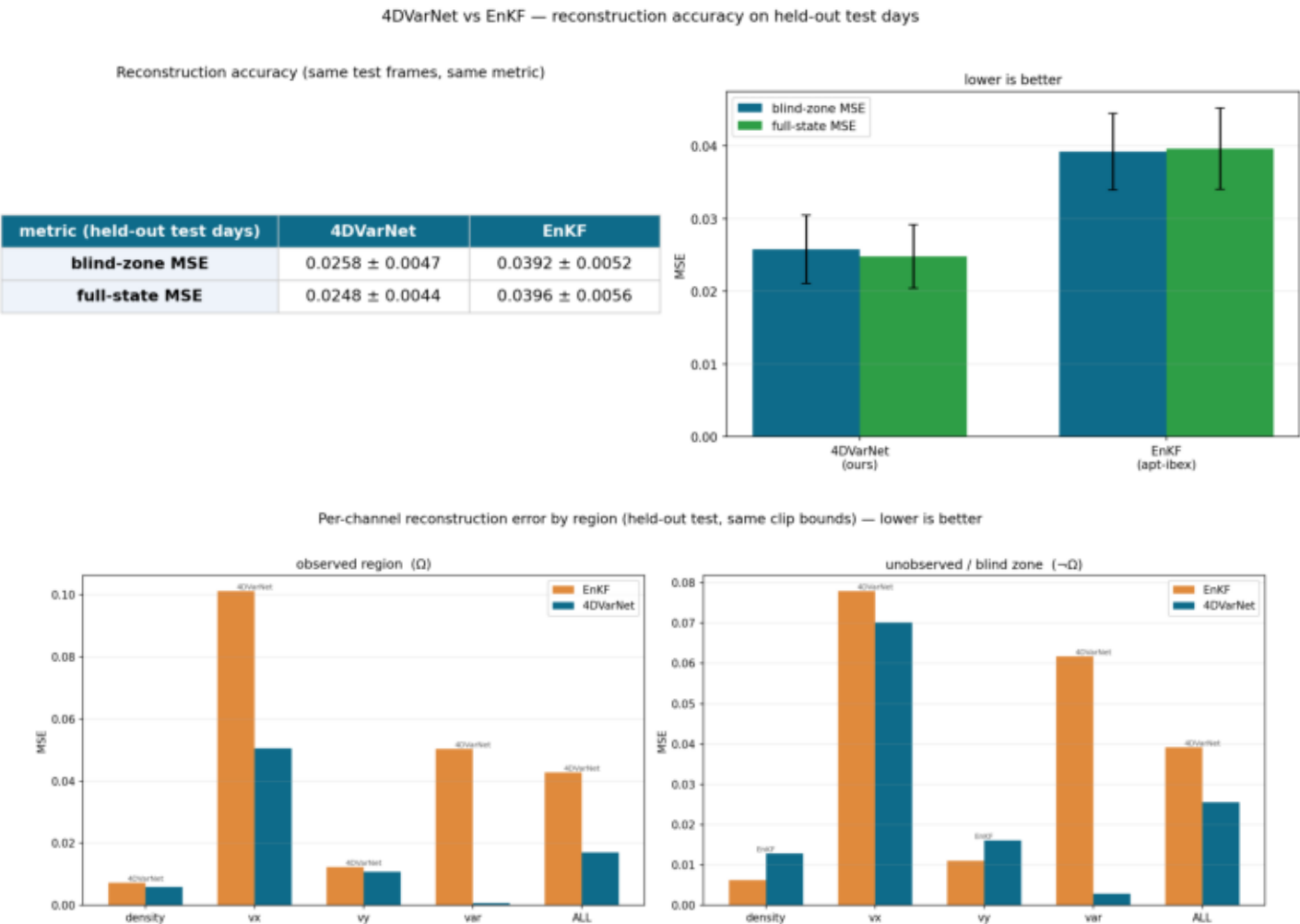
- both methods evaluated on the SAME 7 held-out test days
- SAME observations (same robot coverage, same additive noise), SAME frames, SAME metric, SAME physical clip bounds
- NEITHER sees the ground truth at initialisation:
 - 4DVarNet starts from the observation-based fill X_0 (paper: solver uses only observations, not true states)
 - EnKF starts from its ORIGINAL random Gaussian ensemble (unchanged from the Partial_observation project)
- comparison is reconstruction ACCURACY only — the EnKF's uncertainty (ensemble spread) is shown separately, not scored against a point estimate
- → any accuracy difference reflects the METHOD, not an unfair setup

5 — Reconstruction: density and velocity (EnKF-baseline style)



colour: top = density (+ EnKF spread), bottom = speed $|v|$ · arrows = heading · panels: obs | true | 4DVarNet | EnKF

6 — Results: 4DVarNet vs EnKF (held-out test days)



- blind-zone MSE: 4DVarNet 0.0258 vs EnKF 0.0392 (4DVarNet better overall) · full-state: 0.0248 vs 0.0396
- honest per-channel (blind zone) — density: EnKF · vx: 4DVarNet · vy: EnKF · var: 4DVarNet · EnKF's var channel collapses in both regions; that is the main gap

Appendix — Architecture in detail (backup)

